

Experiment 15: Write a Java Program to develop a GUI based Calculator application using the Swing Package.

Aim

To design and implement a **Calculator application using Java Swing** that performs basic arithmetic operations such as **addition, subtraction, multiplication, and division** using event-driven programming.

Theory

Java Swing is a part of the **Java Foundation Classes (JFC)** and is used to create **platform-independent Graphical User Interface (GUI)** applications. Swing provides a rich set of lightweight components such as JFrame, JButton, and JTextField that help in building interactive desktop applications.

Algorithm

1. Start the program.
2. Create a JFrame to serve as the calculator window.
3. Add a JTextField to display numbers and results.
4. Create JButton objects for digits (0–9), operators (+, −, ×, ÷), clear (C), and equals (=).
5. Position all components using setBounds() and add them to the frame.
6. Register ActionListener for all buttons.
7. When a number button is clicked, append the digit to the text field.
8. When an operator button is clicked:
 - Store the current value as the first operand.
 - Store the selected operator.
 - Clear the text field.
9. When the equals (=) button is clicked:
 - Read the second operand.
 - Perform the arithmetic operation based on the stored operator.
 - Display the result in the text field.
10. If the clear (C) button is pressed, reset all values and clear the display.

Program Code

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class CalculatorSwing extends JFrame implements ActionListener {

    JTextField t1;
    JButton b1, b2, b3, b4, b5, b6, b7, b8, b9, b0;
    JButton bAdd, bSub, bMul, bDiv, bEq, bClr;

    double num1 = 0, num2 = 0, result = 0;
    char operator;

    CalculatorSwing() {
        setTitle("Calculator");
        setSize(300, 400);
        setLayout(null);

        t1 = new JTextField();
        t1.setBounds(20, 20, 250, 40);
        t1.setEditable(false);
        add(t1);

        b1 = new JButton("1"); b2 = new JButton("2"); b3 = new JButton("3");
        b4 = new JButton("4"); b5 = new JButton("5"); b6 = new JButton("6");
        b7 = new JButton("7"); b8 = new JButton("8"); b9 = new JButton("9");
        b0 = new JButton("0");

        bAdd = new JButton("+");
        bSub = new JButton("-");
        bMul = new JButton("*");
        bDiv = new JButton("/");
        bEq = new JButton("=");
        bClr = new JButton("C");

        JButton[] buttons = {
            b7, b8, b9, bDiv,
            b4, b5, b6, bMul,
            b1, b2, b3, bSub,
            b0, bClr, bEq, bAdd
        };

        int x = 20, y = 80;
```

```

for (int i = 0; i < buttons.length; i++) {
    buttons[i].setBounds(x, y, 55, 40);
    add(buttons[i]);
    buttons[i].addActionListener(this);
    x += 60;
    if ((i + 1) % 4 == 0) {
        x = 20;
        y += 50;
    }
}

setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
setVisible(true);
}

public void actionPerformed(ActionEvent e) {

    if (e.getSource() == bClr) {
        t1.setText("");
        num1 = num2 = result = 0;
    }

    if (e.getSource() == bEq) {
        num2 = Double.parseDouble(t1.getText());

        switch (operator) {
            case '+': result = num1 + num2; break;
            case '-': result = num1 - num2; break;
            case '*': result = num1 * num2; break;
            case '/': result = num1 / num2; break;
        }
        t1.setText(String.valueOf(result));
    }

    if (e.getSource() == bAdd || e.getSource() == bSub ||
        e.getSource() == bMul || e.getSource() == bDiv) {

        num1 = Double.parseDouble(t1.getText());
        operator = e.getActionCommand().charAt(0);
        t1.setText("");
    }

    JButton[] nums = { b0, b1, b2, b3, b4, b5, b6, b7, b8, b9 };
    for (JButton b : nums) {

```

```

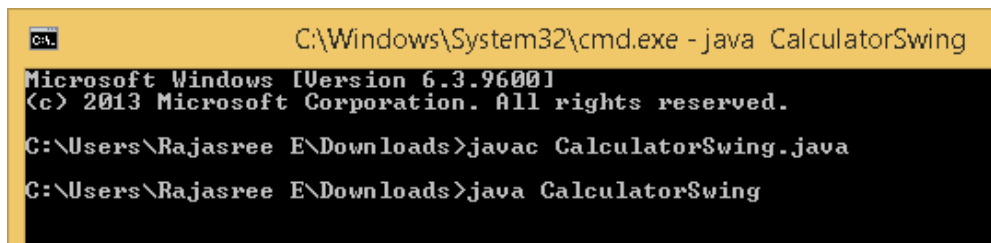
        if (e.getSource() == b) {
            t1.setText(t1.getText() + b.getText());
        }
    }
}

public static void main(String[] args) {
    new CalculatorSwing();
}
}

```

Output:

Command:

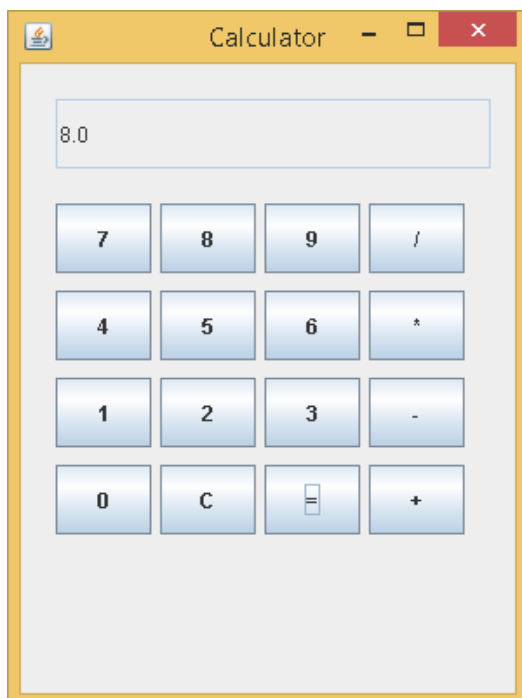


```

C:\Windows\System32\cmd.exe - java CalculatorSwing
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.
C:\Users\Rajasree E\Downloads>javac CalculatorSwing.java
C:\Users\Rajasree E\Downloads>java CalculatorSwing

```

Swing (Interface)



Result

The **Calculator application using Java Swing** was successfully designed and executed. The program correctly performs **addition, subtraction, multiplication, and division** operations based on user input and displays accurate results through a graphical interface. Thus, the objective of implementing a basic calculator using **Java Swing and event handling** was achieved.